

YUJIN HAM

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RESEARCH INTERESTS

My research interests are in the areas of computer vision, with a focus on generative video editing/inpainting, large-scale 3D reconstruction, 3D scene understanding.

EDUCATION

Rice University Ph.D., Department of Electrical and Computer Engineering • Advisor: <i>Prof. Guha Balakrishnan</i> ↗	Houston, TX, United States Aug 2022 – Present
Ewha Womans University M.S., Department of Electronic and Electrical Engineering • Thesis advisor: <i>Prof. Je-Won Kang</i> ↗ • Thesis title: Quality-adaptive Image Compression Artifact Removal using Deep Learning • GPA: 4.23 / 4.30	Seoul, South Korea Mar 2020 – Feb 2022
Ewha Womans University B.S. in Engineering, Department of Electronics Engineering • GPA: 3.54 / 4.30	Seoul, South Korea Mar 2016 – Feb 2020

PUBLICATIONS AND PRESENTATIONS

1. J. Murrugarra-Llerena, P. Chitale, **Y. Ham**, K. Ao, Z. Liu, G. Balakrishnan, P. Cascante-Bonilla, “EgoGroups: A Large-Scale Dataset for Detecting Groups of People in the Wild”, *In Submission*, 2026.
2. **Y. Ham**, J. Kim, V. Boominathan, G. Balakrishnan, “Generating Humanless Environment Walkthroughs from Egocentric Walking Tour Videos”, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
3. **Y. Ham**, M. Michalkiewicz, G. Balakrishnan, “DRAGON: Drone and Ground Gaussian Splatting for 3D Building Reconstruction”, *International Conference on Computational Photography (ICCP)*, 2024. [↗](#)
4. **Y. Ham**, C. Yoo, and J. Kang, “Training compression artifacts reduction network with domain adaptation”, *Applications of Digital Image Processing XLIV. Vol. 11842. International Society for Optics and Photonics*, 2021. [↗](#)
5. (Kor.) **Y. Ham**, C. Yoo, J. Kang, “Compression Artifacts Invariant Training with Domain Adaptation”, *Korean Signal Processing Conference*, 2021.
6. (Kor.) **Y. Ham**, J. Kang, “Mid-view Quantization Noise Removal of Multi-view Video using Convolutional Neural Network”, *The Korean Institute of Broadcast and Media Engineers*, 2020.
7. (Kor.) N. Kim, **Y. Ham**, J. Kang, “Effective Video Captioning Algorithm Using Feature Attention Model”, *Image Processing and Image Understanding*, 2020.

RESEARCH EXPERIENCES

Samsung Research America (SRA) Mentor: <i>Ph.D. Abhinav Kumar</i> ↗ Research Intern, Mobile Processor Innovation (MPI) • keywords: AI Computational Imaging Algorithm	May 2026 – Present
Rice Vision and Imaging Group (RVIG) Advisor: <i>Prof. Guha Balakrishnan</i> ↗ Research Assistant, Rice University • keywords: Video Inpainting, Large scale 3D Reconstruction, 3D Scene Understanding	Aug 2022 – Present
Information Coding and Processing Lab. (ICPL) Advisor: <i>Prof. Je-Won Kang</i> ↗ Research Assistant, Ewha Womans University • keywords: Compression Artifact Reduction, Image enhancement, DA, RL, Blind Image Quality Assessment (BIQA)	Mar 2020 – Feb 2022

Undergraduate Research Assistant, Ewha Womans University	Dec 2018 – Feb 2019
• Subject: Photos classifying Application by exploiting Histogram of Oriented Gradients (HOG)	
Multi-agent Communications and Networking Lab. (MCNL) Advisor: Prof. Hyung-Gon Park ↗	Jun 2018 – Aug 2018
Undergraduate Research Assistant, Ewha Womans University	
• Subject: Convolutional Neural Network (CNN) based classifier and SVM-based classifier performance comparison	

PROJECTS

Walk-through Rendering from Images of Varying Altitude (WRIVA)	Dec 2022 – Present
<i>Intelligence Advanced Research Projects Activity (IARPA)</i> ↗	
• Keywords: Multi-elevation 3D reconstruction, Large scene 3D reconstruction • Improved reconstruction quality by introducing perceptual regularizer using DreamSim • Enhanced rendering quality using Difx3D with scene type-matched reference images across varying altitudes	
Compression and Transmission Technologies for Ultra High Quality Immersive Videos Supporting 6DoF	Mar 2020 – Dec 2020
<i>Institute of Information & Communications Technology Planning & Evaluation (IITP)</i> ↗	
• Keywords: DL-based image denoising, 6DoF data, Quantization noise removal	
A Convolutional Neural Network based Classification System for Educational Learning States using Pupil Sizes	Mar 2018 – Dec 2018
<i>Korea Center for Women in Science, Engineering, and Technology (WISET)</i> ↗	
• Keywords: Machine Learning (ML), Biomedical data classification	
	Seoul, South Korea

TEACHING EXPERIENCES

SWITCH Graduate Mentor Rice University	
<i>Summer Web-Based Institute for Technologies in CompSci and Healthcare (SWITCH)</i> ↗	
• Instructor: Prof. Ashutosh Sabharwal	
Teaching Assistant	
<i>ELEC 542: Generative AI for Image Data (Rice University)</i>	Fall 2026
• Instructor: Prof. Guha Balakrishnan	
<i>ELEC 542: Neural methods for image synthesis (Rice University)</i>	Fall 2023
• Instructor: Prof. Guha Balakrishnan	
<i>30266-01: Digital Engineering (Ewha Womans University)</i>	Spring 2021
• Instructor: Prof. Su-Hyun Park	
<i>35477-01: Random Process (Ewha Womans University)</i>	Fall 2020
• Instructor: Prof. Nak-Myeong Kim	
Undergraduate Peer Instructor Ewha Womans University	Spring 2018 / Fall 2019
Instructor: Prof. Hye-Sook Lim (<i>Digital Engineering</i>) / Prof. Hyung-Gon Park (<i>Random Process</i>)	

HONORS AND AWARDS

Ignite Entrepreneurship Trek to Silicon Valley ↗ Rice University	2025
SWITCH Mentoring Scholarship ↗ Rice University	2024
ICCP 2024 Student Travel Award US National Science Foundation (NSF) & Swiss NSF	2024
Rice University Department of Electrical and Computer Engineering Fellowship Rice University	2022
Grace Hopper Celebration (GHC) Scholarship 2021 AnitaB.org ↗ & Association for Computing Machinery (ACM)	2021
Admission Scholarship for Outstanding Scientists (full tuition for one year) Ewha Womans University	2020
Student Portfolio Award (1st prize) Accreditation Board for Engineering Education of Korea (ABEEK)	2019
Ewha Career Design Scholarship (Merit-based) Ewha Womans University	2019
Dean's List Ewha Womans University	Spring / Fall 2019
Engineering Leadership Scholarship (Merit-based) Ewha Womans University	2018, 2019
Peer Instructor Scholarship (Merit-based) Ewha Womans University	Spring 2018, Fall 2019
Global Frontier Travel Grant (Visiting the US for 2 weeks) Ewha Womans University	2018

SKILLS

Languages: English(fluent), Korean(native)

Programming: Python, C++ , LaTeX, VHDL/Verilog(basic), ARM Assembly(basic), OrCAD (with PSPICE)

ML/DL Frameworks: PyTorch, TensorFlow, MATLAB

OS/Environments: Linux, Mac, Windows, Anaconda, Docker